LESSON 145 (0.5) Ground (1.5) Local Dual (0.3) Instrument

Lesson Objective

The instructor will review and evaluate the applicant's proficiency to determine performance areas which need additional practice prior to course completion.

Pre-Brief

- Review Maneuvers and Procedures to be Performed in Flight.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- Student questions.

Instructor Guidance

- All maneuvers in this lesson are held to FAA Private Pilot ACS standards in direct preparation for the End-of-Course practical test. Make this expectation clear during the pre-brief — the student should treat this lesson as a rehearsal for the EOC, not as a standard practice flight.
- All maneuvers should be conducted "student says, student does" with minimal procedural correction. Your role in this lesson is evaluator, not instructor — intervene for safety and significant errors, but allow minor deviations to play out so you can accurately assess the student's real-world readiness.
- Ensure pre-maneuver checks are completed before every maneuver without prompting. A student who skips clearing turns or rushes the pre-maneuver check on this lesson will do the same on the EOC — address it directly in the post-brief if it occurs.
- Correct mistakes at the point of occurrence rather than holding all feedback for the debrief. Timely corrections during the flight reinforce correct technique and prevent bad habits from being practiced repeatedly.
- Keep a running note of every maneuver that falls outside ACS tolerances and every procedural step that is missed. The post-brief should give the student a specific, prioritized list of items to address before the EOC — vague feedback is not useful at this stage of training.
- Be direct and honest in the post-brief. The student needs a clear, accurate picture of where they stand relative to ACS standards. If they are not yet ready for the EOC, communicate that clearly and outline exactly what needs to improve before proceeding.

Lesson Content

Checklist Procedures

- Have the student run through all checklists.
- Incorporate flows for memory items.
    - Student should be able to perform all briefings.

Preflight Inspection
- Ensure that the student conducted a thorough preflight inspection.

Operation of the Communications and Navigation Units
- Ensure that the student can proficiently use COMS and NAV units.

Engine Starting
- Correctly determine and use appropriate start procedure.
- Incorporate the use of flows.

Radio Communication Procedures and ATC Phraseology
- The student should be able to conduct radio calls at towered and non-towered airports.
- The student should be able to conduct radio calls in the practice area.

Visual Scanning and Collision Avoidance
- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!
- Ensure correct right-of-way rules are being used.

Taxiing
- The student should be able to taxi on the centerline at appropriate speeds on the ramp and taxiways without instructor guidance.
- Ensure the correct use of wind correction throughout taxi.

Normal and Crosswind Takeoff
- The student should conduct the takeoff.
- Ensure the runway entry check and the lineup check are completed.
- Ensure proper crosswind correction is maintained.
- Standards:
  - Maintain manufacturer recommended speed or V_Y +10/-5kts

Short-Field Takeoff and Maximum Performance Climb
- Student says, student does:
  - Verify correct procedure.
- Standards:
  - Apply brakes while setting engine power to achieve maximum performance.
  - Confirm takeoff power prior to brake release and verify proper engine and flight instrument indications prior to rotation
  - Rotate and lift off at the recommended airspeed and accelerate to the recommended obstacle clearance airspeed or VX, +10/-5 knots.
  - Maintain manufacturer recommended speed or V_X +10/-5kts until obstacle is cleared.

- Establish V_Y +10/-5kts, configure plane in accordance to manufacturers guidance after positive rate is verified.
- Maintain VY +10/-5 knots to a safe maneuvering altitude.

Soft-Field Takeoff and Climb

- Student says, student does:
  - Verify correct procedure.
- Standards:
  - Clear the area, maintain necessary flight control inputs, taxi into takeoff position and align the airplane on the runway centerline without stopping, while advancing the throttle smoothly to takeoff power.
  - Establish and maintain a pitch attitude that transfers the weight of the airplane from the wheels to the wings as rapidly as possible.
  - Lift off at the lowest possible airspeed and remain in ground effect while accelerating to VX or VY , as appropriate.
  - Maintain V_X or V_Y , as appropriate, +10/-5 knots to a safe maneuvering altitude.

Steep Turns

- Student says, student does.
  - Verify correct procedure.
- Standards:
  - Roll into a coordinated 360° steep turn with approximately a 45° bank.
  - Perform the task in the opposite direction.
  - Maintain the entry altitude ±100ft.
  - Airspeed ±10kts.
  - Bank ±5°.
  - Roll out on the entry heading ±10°.

Maneuvering During Slow Flight

- Student says, student does.
  - Verify correct procedure.
- Standards:
  - Maintain altitude ±100ft.
  - Airspeed +10kts/-0kts.
  - Heading ±10°.
  - Specified bank ± 10°.

Power-Off Stalls

- Student says, student does.
  - Verify correct procedure.
- Standards:
  - Maintain a specified heading ±10° if in straight flight

    - Maintain a specified angle of bank not to exceed 20°, ±10° if in turning flight, while inducing the stall.

Power-On Stalls

- Student says, student does.
    - Verify correct procedure.
- Standards:
    - Maintain a specified heading ±10° if in straight flight
    - Maintain a specified angle of bank not to exceed 20°, ±10° if in turning flight, while inducing the stall.

Emergency Descent

- Simulate a situation that would require an emergency descent (e.g., smoke or engine fire).
- Student says, student does.
    - Verify correct procedures.
- Standards:
    - Use bank angle between 30° and 40° to maintain positive load factors during the descent.
    - Maintain appropriate airspeed +0/-10kts and level off at specified altitude ±100ft.

Emergency Approach and Landing (Simulated)

- Simulate an engine failure in flight:
    - Student says, student does.
        - Verify correct procedure.
        - Evaluate landing point.
        - Landing must be made with a head wind.
        - Ensure the student verifies with checklists.
- Standards:
    - Establish and maintain the recommended best glide airspeed, ±10 knots.
    - Select a suitable landing area considering altitude, wind, terrain, obstructions, and available glide distance.
    - Plan and follow a flightpath to the selected landing area considering altitude, wind, terrain, and obstructions.
    - Complete the appropriate checklist.

Systems and Equipment Malfunctions

- Simulate a system or equipment malfunction.
    - Assess student for positive aircraft control, decision making skills, and startle response.
    - Verify student uses the appropriate checklist(s) and procedures.
    - Ensure student uses the appropriate action.
- Options for simulation:
    - Causes of partial or complete power loss related to the specific type of powerplant(s).

- System and equipment malfunctions specific to the aircraft, including:
  - Electrical malfunction
  - Vacuum/pressure and associated flight instrument malfunctions
  - Pitot-static system malfunction
  - Electronic flight deck display malfunction
  - Landing gear or flap malfunction
  - Inoperative trim
- Causes and remedies for smoke or fire onboard the aircraft.
- Any other system specific to the aircraft (e.g., supplemental oxygen, deicing).
- Inadvertent door or window opening.

Ground Reference Maneuvers

- Rectangular Course
  - Note: this can be assessed while doing traffic patterns.
  - Student says, student does.
    - Verify correct procedure.
    - Verify proper wind correction on the various pattern legs.
  - Standards:
    - Enter a left or right pattern, 600 to 1,000 feet above ground level (AGL) at an appropriate distance from the selected reference area, 45° to the downwind leg.
    - Apply adequate wind-drift correction during straight and turning flight to maintain a constant ground track around a rectangular reference area, or to maintain a constant radius turn on each side of a selected reference line or point.
    - Maintain altitude ±100 feet; maintain airspeed ±10 knots.
- S-Turns
  - Student says, student does.
    - Verify correct procedure.
    - Both turns must have the same radius.
    - Bank must be adjusted for ground speed/wind.
  - Standards:
    - Enter perpendicular to the selected reference line, 600 to 1,000 feet AGL at an appropriate distance from the selected reference area.
    - Apply adequate wind-drift correction during straight and turning flight to maintain a constant ground track around a rectangular reference area, or to maintain a constant radius turn on each side of a selected reference line or point.
    - Maintain altitude ±100 feet; maintain airspeed ±10 knots.
- Turns Around a Point
  - Student says, student does.
    - Verify correct procedure.
    - Radius must be the same throughout the turn.
    - Bank must be adjusted for ground speed/wind.
  - Standards:

        - Enter at an appropriate distance from the reference point, 600 to 1,000 feet AGL at an appropriate distance from the selected reference area.
        - Apply adequate wind-drift correction during straight and turning flight to maintain a constant ground track around a rectangular reference area, or to maintain a constant radius turn on each side of a selected reference line or point.
        - Maintain altitude ±100 feet; maintain airspeed ±10 knots.

Basic Instrument Maneuvers

- The student must be able to perform the following:
    - Straight-and-level flight.
    - Constant airspeed climbs and descents.
    - Turns to headings.
    - Recovery from unusual attitudes.
- Standards:
    - Maintain altitude ±200 feet, heading ±20°, and airspeed ±10 knots.
    - Level off at the assigned altitude and maintain altitude ±200 feet, heading ±20°, and airspeed ±10 knots.
    - Turn to headings, maintain altitude ±200 feet, maintain a standard rate turn, roll out on the assigned heading ±10°, and maintain airspeed ±10 knots.
    - Recognize unusual flight attitudes; perform the correct, coordinated, and smooth flight control application to resolve unusual pitch and bank attitudes while staying within the airplane's limitations and flight parameters.

Traffic Patterns

- Student should be able to fly traffic patterns and demonstrate understanding of;
    - Towered and nontowered airport operations.
    - Traffic pattern selection for the current conditions.
    - Right-of-way rules.
    - Use of automated weather and airport information.
- Standards:
    - Identify and interpret airport/seaplane base runways, taxiways, markings, signs, and lighting.
    - Comply with recommended traffic pattern procedures.
    - Correct for wind drift to maintain the proper ground track.
    - Maintain orientation with the runway/landing area in use.
    - Maintain traffic pattern altitude, ±100 feet, and the appropriate airspeed, ±10 knots.
    - Maintain situational awareness and proper spacing from other aircraft in the traffic pattern.

Forward Slip to a Landing

- Student says, student does.
    - Verify correct procedure.

  - Power must be at idle.
  - Ailerons into the wind.
  - Rudder opposite.
- Standards:
  - Configure the airplane correctly.
  - As necessary, correlate crosswind with direction of forward slip and transition to sideslip before touchdown.
  - Touch down at a proper pitch attitude, within 400 feet beyond or on the specified point, with no side drift, and with the airplane's longitudinal axis aligned with and over the runway center/landing path.
  - Maintain a ground track aligned with the runway center/landing path.

Go-Arounds or Rejected Landings

- Have your student practice go-around procedures.
- Encourage your student to call for a go-around themselves rather than waiting for you to call it.
- If you do call for a go-around, the student should immediately execute the procedure rather than discussing it first.

Normal and Crosswind Landing

- Student must conduct a normal/crosswind landing.
- Note: If a crosswind condition does not exist, the applicant's knowledge of crosswind elements must be evaluated through oral testing.
- Standards:
  - Select and aim for a suitable touchdown point considering the wind conditions, landing surface, and obstructions
  - Establish the recommended approach and landing configuration, airspeed, and trim, and adjust pitch attitude and power as required to maintain a stabilized approach.
  - Maintain manufacturer's published approach airspeed or in its absence not more than 1.3 times the stalling speed or the minimum steady flight speed in the landing configuration (VSO), +10/-5 knots with gust factor applied.
  - Maintain directional control and appropriate crosswind correction throughout the approach and landing.
  - Make smooth, timely, and correct control application during round out and touchdown
  - Touch down at a proper pitch attitude, within 400 feet beyond or on the specified point, with no side drift, and with the airplane's longitudinal axis aligned with and over the runway center/landing path.
  - Execute a timely go-around if the approach cannot be made within the tolerances specified above or for any other condition that may result in an unsafe approach or landing.

Short-Field Approach and Landing

- Student must conduct a short-field landing.
  - Student does, student says:
  - Verify correct procedure.
  - Emphasize the importance of not forcing the airplane on the ground. If the point cannot be made, perform a go-around.
  - Verify simulated maximum breaking.
- Standards:
  - Maintain manufacturer's published approach airspeed or in its absence not more than 1.3 VSO, +10/-5 knots with gust factor applied.
  - Maintain directional control and appropriate crosswind correction throughout the approach and landing.
  - Make smooth, timely, and correct control application before, during, and after touchdown.
  - Touch down at a proper pitch attitude within 200 feet beyond or on the specified point, threshold markings, or runway numbers, with no side drift, minimum float, and with the airplane's longitudinal axis aligned with and over the runway centerline.

Soft-Field Approach and Landing

- Student must conduct a soft-field landing.
  - Student does, student says:
  - Verify correct procedure.
  - Verify minimal braking.
- Standards:
  - Maintain manufacturer's published approach airspeed or in its absence not more than 1.3 VSO, +10/-5 knots with gust factor applied.
  - Maintain directional control and appropriate crosswind correction throughout the approach and landing
  - Make smooth, timely, and correct control inputs during the round out and touchdown, and, for tricycle gear airplanes, keep the nose wheel off the surface until loss of elevator effectiveness.
  - Touch down at a proper pitch attitude with minimum sink rate, no side drift, and with the airplane's longitudinal axis aligned with the center of the runway
  - Maintain elevator as recommended by manufacturer during rollout and exit the "soft" area at a speed that would preclude sinking into the surface.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

<u>Post Brief</u>

- Student self-critique.
- Instructor critique.

- Student questions.
- Lesson grading.
- Logbook Completion.
- Schedule and preview next lesson.
- Assign Homework:
    - Review missed items from lesson 144 and continuing studying for EOC.

Completion Standards

This lesson is complete when the applicant can:

1. Perform all takeoffs, go-arounds or rejected landings and slips to the FAA Private Pilot ACS.
2. Perform all landings while making smooth, timely, and correct control application during the touchdown and rollout.
3. During all landings:
    a. Make smooth, timely, and correct control application during the touchdown and rollout.
    b. Keep the runway centerline between the main gears of the aircraft.
    c. Touch down smoothly with no drift while keeping the airplane's longitudinal axis aligned with the runway centerline.
4. During normal landings:
    a. Touch down at or within 400 ft beyond a specified point.
5. During short-field landings:
    a. Touch down at or within 200 ft beyond a specified point.
6. During soft-field landings:
    a. Touch down softly with no drift.
7. Perform steep turns while maintaining altitude ±100 ft, airspeed ±10 knots, bank angle ±5° and roll out on the entry heading ±10 degrees.
8. Exhibit the knowledge and skill necessary to perform all checklists and duties related to emergency operations.
9. Perform ground reference maneuvers while maintaining altitude ±100 ft and airspeed ±10 knots.
10. Perform power-off stalls, power-on stalls, and slow flight while maintaining altitude ±100 ft, airspeed ±10 knots and heading ±10 degrees.
11. Perform all items in the basic instrument maneuvers area of operation to FAA Private Pilot ACS.